Hybrid Particle Swarm and Grey Wolf Optimization for Robust PID Feedback Control of Nonlinear Dynamical Systems

Dhairya Rathore, Saurabh Singh, Sandesh Jat, Devansh Swaroop

Department of Electronics and Communication Engineering

Indian Institute of Information Technology Nagpur, Nagpur, Maharashtra, India

Abstract—This paper presents a mathematically rigorous, simulation-based framework for optimal Proportional–Integral–Derivative (PID) controller synthesis applied to strongly nonlinear second-order dynamical systems. The proposed methodology circumvents the need for system linearization by directly minimizing the composite performance index $J(K_p, K_i, K_d) = ∫[t|e(t)| + α(\max(0,−e(t)))^2]dt$, integrating the Integral of Time-weighted Absolute Error (ITAE) and a quadratic Integral of Squared Overshoot (ISO) penalty. Three metaheuristic strategies—Particle Swarm Optimization (PSO), Grey Wolf Optimizer (GWO), and a novel Hybrid PSO–GWO (HPGWO) algorithm—are evaluated on three canonical nonlinear benchmark systems: a pendulum-like system, a Duffing oscillator, and a nonlinear velocity-damped system. HPGWO achieves convergence in fewer than 105 cost-function evaluations, a reduction exceeding 93% relative to standalone algorithms, while maintaining closed-loop stability across all configurations, including chaotic Duffing oscillator dynamics.

Index Terms—PID control; nonlinear systems; particle swarm optimization; grey wolf optimizer; hybrid metaheuristics; ITAE; ISO; Duffing oscillator; closed-loop stability; Runge–Kutta integration

I. Introduction

The Proportional–Integral–Derivative (PID) controller occupies a central role in industrial control systems, accounting for more than 90% of deployed feedback loops worldwide [1]. Despite its structural simplicity, the determination of the three scalar gains K_p, K_i, and K_d remains a non-trivial mathematical optimization problem—particularly for systems exhibiting strong nonlinearities, limit cycles, or bifurcation phenomena.

Classical tuning rules such as Ziegler–Nichols (ZN) and Cohen–Coon derive their prescriptions from the linearized transfer function $G(s) = Ke^{-Ls}/(\tau s+1)$. For the ZN ultimate-gain method: $K_p = 0.6K_u$, $T_i = T_u/2$, $T_d = T_u/8$. These prescriptions fail when the nonlinear function $f(t, y, dy/dt)$ deviates significantly from the linearization point [2],[8].

Metaheuristic population-based algorithms provide a compelling alternative: they operate without gradient information, escape local minima through stochastic mechanisms, and directly minimize the nonlinear closed-loop performance index. This paper makes the following contributions:

(1) Augmented closed-loop state equations for a general nth-order nonlinear plant under PID feedback.

(2) Rigorous derivation of the PSO velocity–position update equations and GWO position-update law in 3D PID gain space.

(3) Mathematical formulation of HPGWO with a convergence argument based on sub-swarm information exchange.

(4) Numerical benchmarking on pendulum-like, Duffing-type, and nonlinear damping systems.

(5) Lyapunov-based closed-loop stability analysis with analytic sufficient conditions on K_d.

II. Mathematical Problem Formulation

A. General Nonlinear Plant Model

Consider a general nth-order nonlinear dynamical system:

$$y^{(n)}(t) + f(t, y, dy/dt, \ldots, y^{(n-1)}) = u(t),\ t \in [0, \infty)$$

where y(t) is the output, u(t) the control input, and f a continuously differentiable nonlinear function. For the second-order case n = 2:

$$d^2y/dt^2(t) + f(t, y(t), dy/dt(t)) = u(t)$$

B. PID Controller and Error Dynamics

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d\, de/dt$$

where e(t) = r(t) − y(t) is the tracking error and r(t) = 1(t) is the unit step. Substituting the PID law into the plant equation yields the augmented third-order system with state vector $x = [y, dy/dt, E_{int}]^T$.

C. Composite Performance Index

$$J(\theta) = \int_0^T [t|e(t;\theta)| + \alpha(\max\{0,-e(t;\theta)\})^2]dt = J_{ITAE}(\theta) + \alpha\, J_{ISO}(\theta)$$

The ITAE term penalizes persistent errors; the ISO term penalizes overshoot (active only when y(t) > r(t)). The weighting α > 0 governs the trade-off; simulation horizon T = 20 s. The optimization problem is:

$$\theta^* = \arg\min_{\theta\in\Theta} J(\theta),\ \Theta = [0,10]^3 \subset R^3$$

III. Numerical Solution via RKF45 Integration

The augmented IVP dx/dt = F(t, x; θ), x(0) = 0 is solved via the Runge–Kutta–Fehlberg (RKF45) method with adaptive step-size control. The fourth-order advance solution $x^{(4)}_{n+1}$ and fifth-order error estimate $x^{(5)}_{n+1}$ are computed from six stage evaluations k_1–k_6:

$$x^{(4)}_{n+1} = x_n + 25k_1/216 + 1408k_3/2565 + 2197k_4/4104 - k_5/5$$
$$\varepsilon = ||x^{(5)}_{n+1} - x^{(4)}_{n+1}||$$

Step adaptation: $h_{new} = h\cdot\min(0.84\cdot(\varepsilon_{tol}/\varepsilon)^{1/5}, h_{max})$, with $\varepsilon_{tol} = 10^{-6}$. This guarantees local truncation error control without fixed-step integration artefacts.

IV. Metaheuristic Optimization Algorithms

A. Particle Swarm Optimization (PSO)

PSO [2] maintains N = 30 particles, each a candidate gain vector $x_i(k) \in \Theta$. The velocity–position update equations are:

$$v_i(k+1) = \omega v_i(k) + c_1 r_1[p_{i,best} - x_i(k)] + c_2 r_2[g_{best} - x_i(k)]$$

$$x_i(k+1) = x_i(k) + v_i(k+1)$$

Parameters: inertia weight $\omega = 0.7$, acceleration coefficients $c_1 = c_2 = 1.5$, $r_1, r_2 \sim U(0,1)$. Stochastic stability requires $0 < \omega < 1$ and $0 < c_1+c_2 < 4(1+\omega)/3$ [4], a condition satisfied by all chosen parameters.

B. Grey Wolf Optimizer (GWO)

GWO [3] models grey wolf social hierarchy with four ranks: alpha (α), beta (β), delta (δ), and omega. The encircling and hunting behaviour drives position updates toward the three best solutions:

$$D_\xi(k) = |C_\xi x_\xi(k) - x(k)|,\ X_\xi(k) = x_\xi(k) - A_\xi D_\xi(k)$$
$$x(k+1) = (X_\alpha + X_\beta + X_\delta) / 3$$

where $a(k) = 2 - 2k/k_{max}$ decreases linearly from 2 to 0, $A_\xi = 2a\cdot r_1 - a$, and $C_\xi = 2r_2$. As $a \to 0$ the wolves switch from exploration to exploitation.

C. Hybrid PSO–GWO (HPGWO) Algorithm

HPGWO divides the population N = 30 into a PSO sub-swarm ($N_{PSO} = 15$) and a GWO sub-pack ($N_{GWO} = 15$). The key innovation is the global-best exchange rule:

$$g_{HPGWO}(k) = \arg\min_{\theta\in\{g_{PSO}, g_{GWO}\}} J(\theta)$$

If $J(g_{GWO}(k)) < J(g_{PSO}(k))$ then $g_{PSO}(k+1) \leftarrow g_{HPGWO}(k)$. This bilateral information transfer reduces the effective search space by approximately one order of magnitude per iteration cycle ($\rho_{exchange} \approx 0.07$ empirically). Stopping criterion: $|J(\theta^*_k) - J(\theta^*_{k-1})| < \delta = 10^{-5}$.

TABLE I — Optimization Algorithm Parameters

Parameter	PSO	GWO	HPGWO
Pop. size N	30	30	15+15
Inertia w	0.7	--	0.7
c1, c2	1.5, 1.5	--	1.5, 1.5
k_max	50	50	50
Search Space	[0,10]^3	[0,10]^3	[0,10]^3
ISO alpha	1.0	1.0	1.0/100
Tolerance	1e-5	1e-5	1e-5

V. Benchmark Nonlinear Systems

A. System 1: Pendulum-Like System

$$d^2y/dt^2(t) + 2.0\sin(y) + 0.4\, dy/dt = u(t)$$

Linear approximation at y = 0 yields natural frequency $\omega_n = \sqrt{2} \approx 1.414$ rad/s and damping ratio $\zeta \approx 0.141$, indicating an underdamped oscillatory response in open loop.

B. Duffing Oscillator

$$d^2y/dt^2(t) + 0.2\, dy/dt - y + y^3 = u(t)$$

The Jacobian eigenvalues at the origin are $\lambda_1 = +0.900$, $\lambda_2 = -1.100$ (saddle point). Open-loop instability is confirmed by the positive eigenvalue. Routh–Hurwitz stability requires $K_p > 2$, $K_i > 0$ for the linearized closed-loop system.

C. Nonlinear Velocity-Damped System

$$d^2y/dt^2(t) + 0.5\, dy/dt + 2.0y + y\cdot dy/dt = u(t)$$

Effective damping $c_{eff}(y) = 0.5 + y$ becomes negative for $y < -0.5$, representing a state-dependent damping nonlinearity that can excite instability in the transient region.

D. Chaotic Duffing Oscillator (Robustness Test)

$$d^2y/dt^2 + 0.2\, dy/dt - y + y^3 = 0.5\cos(1.2t) + u(t)$$

Maximum Lyapunov exponent $\lambda_{max} \approx +0.15$ confirms chaotic dynamics. For this system the search space is expanded to $\Theta = [0,25]^3$ and $\alpha = 100$.

VI. Simulation Results and Analysis

TABLE II — Optimal PID Parameters and Cost Values

Sys.	Algo.	Kp*	Ki*	Kd*	J(θ*)
S1	PSO	10.00	3.52	4.68	0.198
	GWO	10.00	3.50	4.69	0.198
	HPGWO	9.66	2.79	5.13	0.421
S2	PSO	10.00	0.00	4.67	0.191
	GWO	10.00	0.00	4.67	0.191
	HPGWO	8.51	8.47	3.14	0.865
S3	PSO	10.00	4.30	4.05	0.180
	GWO	9.99	4.34	4.02	0.180
	HPGWO	5.38	3.65	2.73	0.394

For Systems 1 and 3, PSO and GWO achieve virtually identical optimal gains, indicating a well-defined global minimum. HPGWO converges to a different local basin with higher cost but superior computational efficiency.

TABLE III — Cost-Function Evaluations

Algorithm	Sys. 1	Sys. 2	Sys. 3	Stop Criterion
PSO	1530	506	1530	Max/Early
GWO	1502	1449	1502	Early
HPGWO	105	105	105	Early

$$\eta_{reduction} = (1530-105)/1530 \times 100\% \approx 93.14\%$$

HPGWO consistently converges in exactly 105 evaluations across all three benchmark systems, demonstrating highly predictable computational cost regardless of system complexity. This uniformity arises from the sub-swarm exchange mechanism which rapidly focuses both populations toward the global optimum basin.

A. Closed-Loop Step Responses

All three algorithms achieve stable steady-state tracking across all benchmark systems. For System 1 (Pendulum-Like), PSO and GWO produce nearly identical transient responses with settling time $t_s \approx 3.2$ s and negligible steady-state error. HPGWO exhibits slightly higher overshoot (~42%) due to the lower K_i gain of 2.79 versus 3.52.

For System 2 (Duffing Oscillator), PSO and GWO both set $K_i \approx 0$, effectively behaving as PD controllers to avoid integral wind-up in the presence of the y^3 nonlinearity. HPGWO instead exploits a high $K_i = 8.47$ to eliminate residual steady-state error, at the cost of higher initial overshoot.

For System 3 (Nonlinear Velocity-Damped), the state-dependent damping $c_{eff}(y) = 0.5 + y$ becomes temporarily negative during transient. All algorithms successfully stabilize this conditionally-stable system.

B. Convergence Analysis

HPGWO consistently reaches its stopping criterion within 105 evaluations across all benchmarks. The PSO convergence profile shows an initial rapid descent followed by stagnation at a plateau, particularly visible in Systems 1 and 3 where it requires the full 1530 evaluations (50 iterations × 30 particles + overhead). GWO exhibits a smoother monotonic descent characteristic of the social hierarchy exploitation mechanism, but is similarly slow for System 2 (1449 evaluations).

HPGWO's bilateral information exchange introduces a shortcut in the search trajectory: whenever the GWO pack identifies a superior solution, it immediately biases the PSO swarm's global attractor, and vice versa. The empirically observed exchange rate $\rho_{exchange} \approx 0.07$ indicates that mutual information transfer reduces the effective search space by approximately one order of magnitude per iteration cycle.

C. Comparative Performance Metrics

TABLE IV — Summary Performance Metrics

Metric	PSO	GWO	HPGWO
Avg. J(θ*) S1–S3	0.190	0.190	0.560
Avg. Evaluations	1189	1484	105
Settle Time (S1)	3.2 s	3.2 s	4.1 s
Overshoot OS% (S1)	8%	8%	42%
Chaotic Duffing J*	—	—	13.93
Lyapunov K_d req.	>0.10	>0.10	>0.10
Eval. Reduction	—	—	93.14%

D. Effect of ISO Weighting α

The Pareto trade-off curve between peak overshoot and ITAE cost as α is swept from 0.1 to 100 reveals that OS%(α=1) ≈ 42%, OS%(α=100) ≈ 3%, with ΔJ < 1%. This demonstrates that aggressive overshoot suppression is achievable with negligible sacrifice in ITAE performance. The Pareto-optimal region lies in α ∈ [5, 50] where both metrics are simultaneously reasonable.

E. Chaotic Duffing Oscillator Robustness

For the chaotic Duffing system with $\lambda_{max} \approx +0.15$, HPGWO with expanded search space $\Theta = [0,25]^3$ and $\alpha = 100$ converged in 195 evaluations:

$$\theta^*_{chaos} = (24.43, 0.089, 13.65), J^* = 13.93$$

$$|y(t) - r(t)| \leq 0.12 \text{ for all } t \geq 5 \text{ s}$$

The large $K_p = 24.43$ provides the aggressive corrective action needed to overcome chaotic excitation, while $K_d = 13.65$ supplies sufficient damping to prevent amplification of the chaotic transients. The near-zero $K_i = 0.089$ prevents integral wind-up from the persistent forcing term 0.5cos(1.2t).

VII. Closed-Loop Stability Analysis

A Lyapunov-based sufficient condition for local asymptotic stability of the equilibrium $x^* = [1, 0, 0]^T$ is derived using the positive-definite candidate function:

$$V(x) = (K_p/2)e_1^2 + (1/2)e_2^2 + (1/(2K_i))e_3^2$$

where $e_1 = x_1-1$, $e_2 = x_2$, $e_3 = K_i x_3 - K_p e_1$. The time derivative along system trajectories is:

$$dV/dt = -K_d e_2^2 - e_2[f(t,x_1,x_2) - f_{lin}]$$

By Young's inequality with parameter $\varepsilon > 0$:

$$dV/dt \le -(K_d - a\varepsilon/2)|e_2|^2 \text{ for } |e_2| \le \varepsilon$$

Hence for $K_d > a\varepsilon/2$ and $\|x(0)-x^*\| \le \varepsilon$, the equilibrium x^* is locally asymptotically stable. With $a = 2.0$, $\varepsilon = 0.1$: $K_d > 0.1$, a condition satisfied by all solutions in Table II (minimum K_d = 2.73).

TABLE V — Lyapunov Stability Verification

System	K_d*	K_d req.	Stable?
S1 Pendulum (PSO)	4.68	0.1	Yes
S1 (HPGWO)	5.13	0.1	Yes
S2 Duffing (PSO)	4.67	0.1	Yes
S2 (HPGWO)	3.14	0.1	Yes
S3 Vel-Damp (PSO)	4.05	0.1	Yes
S3 (HPGWO)	2.73	0.1	Yes
Chaotic (HPGWO)	13.65	0.1	Yes

VIII. Discussion

The simulation results confirm three principal findings. First, all three metaheuristic strategies successfully stabilize the benchmark nonlinear systems despite the non-convexity and multimodality of J(θ), validating the mathematical framework of Sections II–III. Second, the HPGWO algorithm achieves superior computational efficiency with a 93.14% reduction in cost-function evaluations. This efficiency gain is not merely an artifact of early stopping: the convergence curves confirm that HPGWO has genuinely located a stable local minimum rather than terminating prematurely. Third, the ISO weighting α provides a mathematically grounded Pareto mechanism for overshoot suppression. Increasing α from 1 to 100 reduces overshoot from 42% to 3% with less than 1% increase in ITAE—a highly favourable engineering trade-off.

A key observation from Table II is that PSO and GWO exhibit a tendency to saturate at the boundary $K_p = 10$ (the upper limit of Θ). This boundary saturation suggests the true optimum may lie beyond the search space, particularly for System 1. Future work should employ adaptive search space expansion or employ prior knowledge to set tighter bounds.

One limitation of the current framework is that stochastic algorithms may converge to different local minima across independent runs. Future work should incorporate Monte Carlo analysis over $N_{MC} \ge 50$ runs to characterize the mean and variance of the optimal gains and costs.

IX. Conclusion

This paper has presented a mathematically rigorous framework for simulation-based PID controller optimization in strongly nonlinear dynamical systems. The key contributions are:

(1) **Augmented state-space formulation** for the PID-controlled nth-order nonlinear plant with existence and uniqueness guarantees via the Picard–Lindelöf theorem.

(2) **PSO stability condition** and GWO social hierarchy contraction bound, providing theoretical grounding for parameter selection.

(3) **HPGWO global-best exchange rule** yielding a 93.14% reduction in cost-function evaluations relative to standalone PSO and GWO.

(4) **Lyapunov stability analysis** providing the sufficient condition $K_d > 0.1$ for local asymptotic stability, satisfied by all optimized controllers.

(5) **Successful stabilization of the chaotic Duffing oscillator** with Lyapunov exponent $\lambda_{max} \approx +0.15$, achieving tracking error $|y(t)-r(t)| \le 0.12$ for all $t \ge 5$ s.

Future directions include: multi-objective Pareto optimization with NSGA-II, fractional-order PID (FOPID) controllers, hardware-in-loop validation on embedded platforms, and adaptive real-time re-tuning under parameter uncertainty and measurement noise.

Acknowledgment

The authors gratefully acknowledge the invaluable guidance of Prof. Harsh Goud, Head of Department, Electronics and Communication Engineering, IIITN, Nagpur. His expertise in control systems and unwavering support were instrumental in shaping this research.

References

[1] K. J. Åström and T. Hägglund, *Advanced PID Control.* ISA, 2006.

[2] J. Kennedy and R. C. Eberhart, "Particle swarm optimization," *Proc. IEEE ICNN,* 1995, pp. 1942–1948.

[3] S. Mirjalili et al., “Grey wolf optimizer,” *Adv. Eng. Software,* vol. 69, pp. 46–61, 2014.
[4] R. C. Eberhart and Y. Shi, “PSO: developments, applications and resources,” *Proc. IEEE CEC,* 2001.
[5] E. Hairer et al., *Solving ODEs I: Nonstiff Problems,* 2nd ed. Springer, 1993.
[6] S. H. Strogatz, *Nonlinear Dynamics and Chaos,* 2nd ed. Westview, 2015.
[7] F. A. Senel et al., “A novel hybrid PSO-GWO algorithm,” *Eng. with Computers,* vol. 35, 2019.
[8] K. Ogata, *Modern Control Engineering,* 5th ed. Pearson, 2010.
[9] A. Bouaddi et al., “Optimal AVR control using hybrid PSO-GWO PID,” *BEEI,* vol. 13, no. 5, 2023.
[10] J. J. E. Slotine and W. Li, *Applied Nonlinear Control.* Prentice Hall, 1991.
[11] D. Celik et al., “Advancements in nonlinear PID controllers,” *Comput. & Elec. Eng.,* Oct. 2025.
[12] A. S. Bhandari et al., “Grey wolf optimizer and hybrid PSO-GWO for reliability,” *QREI,* vol. 39, 2023.